Homework Questions for Mind and Brain Week 2

Due the week of September 15

SHOW ALL WORK

Last week you completed two assignments on the Proton adaptive learning platform. In addition to completing another assignment on Proton this week (Lesson 3: Evaluating Scientific Claims), you are also required to answer the following homework questions. You will have such written homework questions in most of the remaining weeks of the semester.

Unlike Proton, which should be completed individually, you may collaborate with peers while working on this assignment. In fact, we encourage you to work with others. However, all submitted answers should be stated in your own words and arrived at from your own understanding. Please do not simply consult AI tools or copy from someone else's assignment – this would constitute academic dishonesty and would be reported as such. In addition to working with peers, your seminar instructor would be happy to help you during their office hours.

The homework questions are designed to help you achieve the course objectives and are an excellent preparation for the course exams. Many of the questions we include come from past exams. Some questions in this homework assignment come from past midterm exams.

1. Studying an Amnesiac's Brain

1A. In this week's lecture, you learned about H.M, an extraordinary patient whose anterograde amnesia fundamentally changed how we understand the relationship between memory and the brain.

Because H.M.'s lesion was due to surgical removal, his brain damage was limited to both of his hippocampi and parahippocampal cortices (part of the medial temporal lobe cortex that is adjacent to the hippocampus). As part of various studies, H.M. performed many different tasks, and researchers found that he had both impaired and spared brain functions.

1A. (i) Briefly describe the impact that H.M.'s surgery had on his memory. Which functions were impaired vs. spared, and what do these findings tell us about memory more generally?

After surgery, H.M. could not make new memories about daily events or new facts. He also forgot some things that happened just before the surgery. His short term memory was still okay. If you told him a number, he could repeat it for a short time. His personality, language, and general thinking were also mostly normal. He could still learn new skills, such as mirror tracing, even though he did not remember practicing. This shows that memory is not one single thing. The hippocampus and nearby areas are needed for making new memories about facts and events. Skill learning uses other parts of the brain and can still work when the hippocampus is damaged.

1A. (ii) Name one advantage and one disadvantage of studying extraordinary individuals like H.M to learn about brain function.

Advantage: studying a person like H.M. helps us see what happens when one part of the brain is damaged. If a person loses a certain type of memory after damage in one area, then maybe this area is important for that type of memory.

Disadvantage: A single case might be special. H.M. was one person with a unique history and injury. What we see in him might not be true for everyone, so it is hard to know how general the results are.

1B. In your reading this week, you also learned about Lonni Sue Johnson. Viral encephalitis damaged her hippocampus and left her with profound amnesia. In one of the experiments, the researchers had Lonni Sue play some musical pieces on the viola.

1B. (i) In 2-3 sentences explain how the experiment was designed and the reasoning behind this design.

The researchers gave Lonni Sue several new viola pieces. She practiced some of them many times over several days. At least one piece was not practiced and was a control. They measured how many notes she played correctly each time. Later, they tested her again on the same pieces. Because she has strong amnesia, she did not remember seeing the pieces before. If she still improved on the practiced pieces, this would show that her skill memory was working, even without memory for the practice sessions.

1B. (ii) What was the researchers' hypothesis?

The researchers thought that Lonni Sue would still get better at playing new viola pieces with practice. They expected skill learning for music to be at least normal even though her memory for events and facts was really damaged.

1B. (iii) What were the results and do they support the hypothesis?

Lonni Sue's playing on the practiced pieces improved a lot. Her accuracy almost doubled. On the piece she didn't practice, there wasn't a lot of improvement. When they tested her later, she still played the practiced pieces better than before, even though she said she didnt remember learning them. This supports the hypothesis. It shows that she could still learn motor skills for music. Skill memory was working, even though her memory for events was not.

2. <u>Diet and memory</u>

In a 2024 study, Hayes and colleagues studied the impact of diet of juvenile rats. They fed only junk food (chips and chocolate) to one group of juvenile rats and a more balanced diet to another group of juvenile rats.

2A. Once the rats reached early adulthood, both groups were given the same balanced diet for a period of time. Then, the scientists measured the memory performance of each group of rats through a series of experiments.

2A. (i) The researchers were interested in junk food's effects on the hippocampus, so they measured how well rats could navigate a room they had seen before. **Suggest a possible hypothesis for this study.**

Rats that eat a junk food diet when they are young will do worse on a memory task as adults. In this study, that means they will have a lower exploration index in a familiar room, compared with rats that had a healthy, balanced diet when they were young. This would still be true even after both groups get the same balanced diet as adults.

2A. (ii) State the null hypothesis for this experiment.

2B. (i) According to the data presented in Figure 1 (next page), would you accept or reject the null hypothesis? Explain using your knowledge of statistics. Answer on next page, below the figure)

<u>[Note: When asked to explain using your knowledge of statistics, you should support your answer with an evaluation of the relevant confidence intervals or consideration of provided p-values. You should state what comparisons you are making, whether the values are or are not statistically significantly different (and how you know: e.g., 95% confidence intervals do or do not overlap, therefore values are statistically significantly different or not), and if statistically significantly different, which value is larger/smaller.]</u>

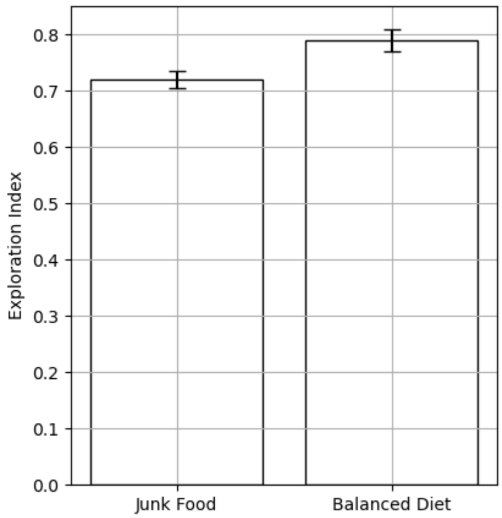

Figure 1. *Exploration index for rats that were fed junk food or a balanced diet in their juvenile stage. The exploration index denotes how well rats navigate a space, where a higher index shows stronger recognition. The error bars represent the 95% confidence intervals.*

2B. (ii) Why might the researchers believe that differences in the exploration index are related to differences in the hippocampi of rats? Explain in one sentence.

2C. After both groups of rats had been fed a balanced diet in their adult stage, the scientists also measured the rats' ability to process glucose (sugar). The reason for this investigation is that if a body cannot process glucose properly, the blood glucose levels increase, and this can lead to diseases such as diabetes.

2C. (i) Why might the scientists have fed both groups of rats a balanced diet in their adult stage before proceeding with the study?

Consider the results shown in Table 1.

Table 1. *Mean and standard deviation (SD) of blood glucose levels for two groups of adult rats.*

Group	**Mean blood glucose level**	**SD blood glucose level**
Given junk food as juveniles N = 16	175 mg/dL	10.2 mg/dL
Given a balanced diet as juveniles N = 16	167 mg/dL	8.6 mg/dL

2C. (ii) Calculate the standard error for the blood glucose level of the two groups.

2C. (iii) Calculate the 95% confidence interval for the blood glucose level of the two groups in units of mg/dL.

2C. (iv) Based on the results presented in Table 1, did the rats that were fed different diets in the juvenile phase have different blood glucose levels as adults? Explain using your knowledge of statistics.

2D. Which of the following options best describe an experiment that could provide converging evidence for the results Hayes and colleagues have obtained? Briefly explain your choice

a. A similar study with double the number of rats.
b. A human study where people answer a survey informing researchers of their eating habits and then perform a memory test.
c. A repetition of the study, where systematic and random errors are identified and fixed.
d. An experiment where rats are trained on a visual discrimination task aiming to characterize their visual acuity.

2E. The authors suggest that the rats' hippocampi are active during the memory test but provide no evidence for that. During the lecture you learned about different techniques for studying the brain. **Choose one of those techniques and explain how it could be used to search for evidence of hippocampal involvement in navigating a maze (for rats) or navigating a new place (for humans). Describe one strength and one limitation of your chosen technique.**